
UNIT 3 MOBILE CLIENT DEVICES AND PERVASIVE COMPUTING

3.0 Introduction

3.1 Objectives

3.2 Smart Sensors, Actuators and Mobile Robotic Systems

3.2.1 Smart Sensors

3.2.2 Actuators

3.2.3 Robotic System Sensors and Actuators

3.3 Smart Home and Appliances

3.3.1 Smart Appliances

3.3.2 Set-top boxes

3.4 Automotive Systems

3.4.1 Speech Recognition System

3.4.2 Messaging System

3.4.3 GPS-based Navigation System

3.4.4 Automobile Start and Malfunction Logins

3.4.5 Sensor and Actuator Programming

3.4.6 Entertainment Systems

3.4.7 Real-time Applications Programming

3.5 Limitations And Devices Design Constraints

3.5.1 Limitations of the Devices

3.5.2 Design Constraints for Handheld Mobile Device Applications

3.6 Summary

3.7 Further Readings

3.0 INTRODUCTION

The client is a mobile application that queries a server and receives a response. The notifications are also sent by the server. A mobile client is a computer that runs Web applications and functions in a client-server context. Hardware and software are both present in the device.

Modern automation technology makes it feasible to control security systems and household appliances from a computer or mobile device.

Numerous miniature computing systems are included in an automotive computing system. Systems for voice control, traffic congestion information, smartcard-based security control, collision avoidance systems, reverse sensing, night vision, and communication to central real-time traffic monitors are a few examples.

Automotive systems have undergone a transformation during the past 10 years or so. The modern automobiles come equipped with everything from sophisticated information-oriented technology like GPS navigation reverse sensing, and night vision to communication systems like e-mail access, voice control, traffic congestion information, smartcard security control, and collision avoidance sensors.

3.1 OBJECTIVES

At the end of this Unit, you shall be able to:

1. Understand the features and capabilities of mobile music players and smartphones.
 2. Understand automotive computing systems.
 3. Understand the types and characteristics of mobile devices.
 4. Understand about the applications and characteristics of set-top boxes, smartcards, labels, sensors, actuators, and home appliances.
 5. Understand the mobile handheld device limitations and design restraints.
-

3.2 SMART SENSORS, ACTUATORS, AND MOBILE ROBOTIC SYSTEMS

In control systems, sensor-actuator pairs are employed. Like, for instance, a temperature sensor and current actuator pair controls the oven's temperature, a light sensor and bulb current actuator pair regulates the brightness, and a pressure sensor and valve actuator pair regulates pressure.

There are a lot of sensor and actuator pairs in industrial plants. A control area network bus (CAN bus), for instance, is used to connect a group of intelligent sensors and actuators in an automotive or industrial facility.

3.2.1 Smart Sensors

Electronic devices called sensors sense the outside world; examples include sensors for temperature, pressure, light, metal, smoke, and object proximity. The controller or computer receives the signals from the sensor. In order to hear voices, a microphone is used.

A sensor could be a complex system of numerous tiny gadgets. A camera with a charge-coupled device (CCD) base is used to detect and recognize different things. The MP range for cameras is 2 to 12. Signals are produced by each pixel of a CCD. The signals are digitally transformed using Analog-to-digital Converter (ADC).

Mobile devices may interact with their surroundings thanks to their sensors. Here are a few instances:

1. Voice amplification during a call can be managed with the aid of a background noise sensor.
2. The LCD screen's brightness can be managed using a sensor for ambient light. When ambient light levels are very low, it minimizes power dissipation for displays.

3. The amplification of received signals is controlled by a sensor that gauges the signal's strength.
4. Voice is detected by a microphone. A speech processing system receives the voice signals from it (SPS). The SPS verifies the cellphone owner's identity. The SPS can then be used to interpret and carry out spoken commands as well as dial a spoken number.
5. A gyroscope tracks the direction and speed of angular motion along three axes. For recording vibration shocks or falls, an accelerometer detects the device's linear acceleration along three axes. A device's power can be turned on using a proximity sensor.

A tiny magnet hung by a wire forms the basis of a smart magnetometer sensor, which connects to an app for data processing and makes intelligent decisions. It detects and measures the values of three magnetic field strength components in the x, y, and z directions. The meter's tiny magnet points in the direction of the magnetic field's force lines.

The lines of force and the strengths of the magnetic fields in the x, y, and z directions all alter when a magnetized material is present nearby. A small magnet's North and South poles are subject to opposing angular forces. This is as a result of shifting field directions and strengths. The wire develops a torque due to the forces. The small magnet is rotated by the torque that the wire experiences. An app analyses the ensuing modifications and notifies the actions in accordance with the sensor's measured data.

The magnetometer and apps are on the latest generation of mobile devices. An app automatically turns off when the user puts their phone in their pocket and turns on when they hold it in their hand.

Sensors having processors and memory are referred to as smart sensors. They are capable of computing, communication, and networking. They are used to transmit data to a network, a controller, or a central computer. Multiple smart sensors are installed in robotic systems and industrial automation systems.

The processor, memory, ADC, signal-processing component, wireless or infrared receiver, and transmitter make up a smart sensor, which executes both computational and communicational tasks. Assembly language or C language is typically used to programme smart sensors.

3.2.2 Actuators

A controller or central computer sends signals to an actuator, which then uses those signals to

activate a physical system, appliance, or device. A servomotor in a robot's hand, a loudspeaker, a power transistor (used to transmit electricity to an oven), a solenoid-valve actuator, a transmitting device in a sensor network, and other physical devices are examples of this type. The physical system or equipment is activated in response to the commands or signals received from a network, mobile device, computer, or controller via a smart actuator.

3.2.3 Robotic System Sensors and Actuators

The sciences of artificial intelligence and mechanical engineering both contribute to the development of robotic systems in various ways. Robotic systems are essentially programmable machines made up of mechanical actuators and sensory organs connected to a computer inside of them. In industrial robotics, the mechanical structure might include manipulators, but in mobility robotics, it might focus on the robot's movement as a vehicle. The following are a few instances of sensors used in robotic systems:

The left and right foot each have acceleration and force sensors.

The hands and head both include infrared distance sensors.

CCD cameras are located in the eyes.

Angular rate sensor in the centre

ears with microphones

shoulders, hands, and head having thermo and touch sensors

The following are some examples of actuators found in robots:

A robot may have a speaker at its lips so that it can speak spoken commands to other robots or communicate sensed data.

Actuators and motors are present at all joints that move, including the hands, wrist, neck, shoulders, feet, and palms of grippers.

The sensors send signals to the embedded processors at the main computer chip of the robot through internal connections. When a group of robots needs to act in unison, the robot wirelessly transmits data to a central server.

Robots that can move and communicate wirelessly are employed in industrial settings to access spaces that are difficult for people to reach. Robot master-slave systems can serve a number of functions. In such a setup, the master robot issues orders to the other (slave) robots.

Figure 1 below depicts a robot's sensors, actuators, and transceivers. Programming for robot sensors can be done in 'C' language or assembly language.

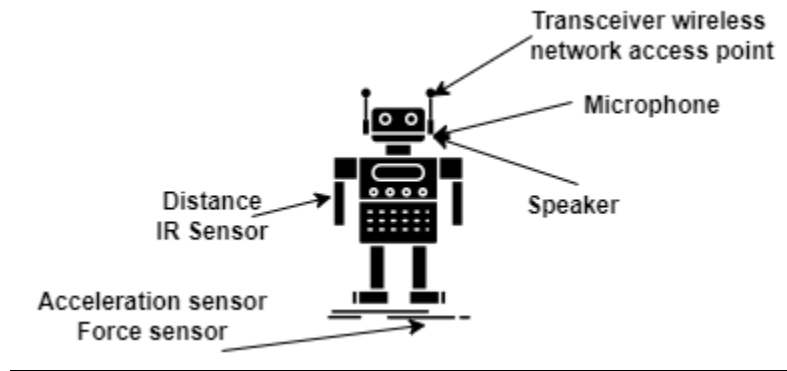


Figure 1

3.3 SMART HOME AND APPLIANCES

Modern automation technology makes it feasible to control security systems and household appliances from a computer or mobile device. Web-enabled gadgets include smart home and workplace products. The following are the networked home devices, a set-top box, and a residential gateway:

3.3.1 Smart Appliances

Using electricity cables, home appliances can be networked. Such wires can induce signals with frequencies as high as 525 kHz. A network can be created by these signals being transmitted from one appliance to another. A central server can be used for the devices to communicate as well. Very short-range wireless protocols like Bluetooth or ZigBee can also be used to network home appliances.

A Web address can be given to a smart appliance. The equipment then utilises a domestic gateway to connect to the Internet. The gateway gives the user access to devices like their home computer, MP3 player, security locks, and other items from the outside via WLAN, the Internet, or a mobile service provider's access point. A network of smart appliances is depicted in Figure.

A domestic gateway is a device that connects your home's electronics to the Internet, including your media player, computer, locks, lights, oven, refrigerator, and air conditioner. The process starts with authentication. The gateway enables access from the outside to the home devices after user authentication.

A service provider server may also be used by the gateway to perform networking tasks.

The mobile service provider may also assign a number to a smart appliance. The SMS service can then be used to control it from a Smartphone. For instance, if a person forgets to turn off a smart air conditioner before leaving the house, it might be turned off by an SMS. Also on the other hand, while returning home, one can turn on the air conditioning by sending an SMS, which will make the room cool before one actually arrives home.

An innovative idea that is currently being developed is electronic appliance maintenance. The appliance can be accessed online by the maintenance service provider, who can then diagnose any problems.

3.3.2 Set-top boxes

A sophisticated computer-based device known as a set-top box connects the home TV to the broadcasting service network. The block diagram of a set-top box's subsystems is shown in Figure 2. There are two sets of communication channels: front and back. This makes interactive TV possible.

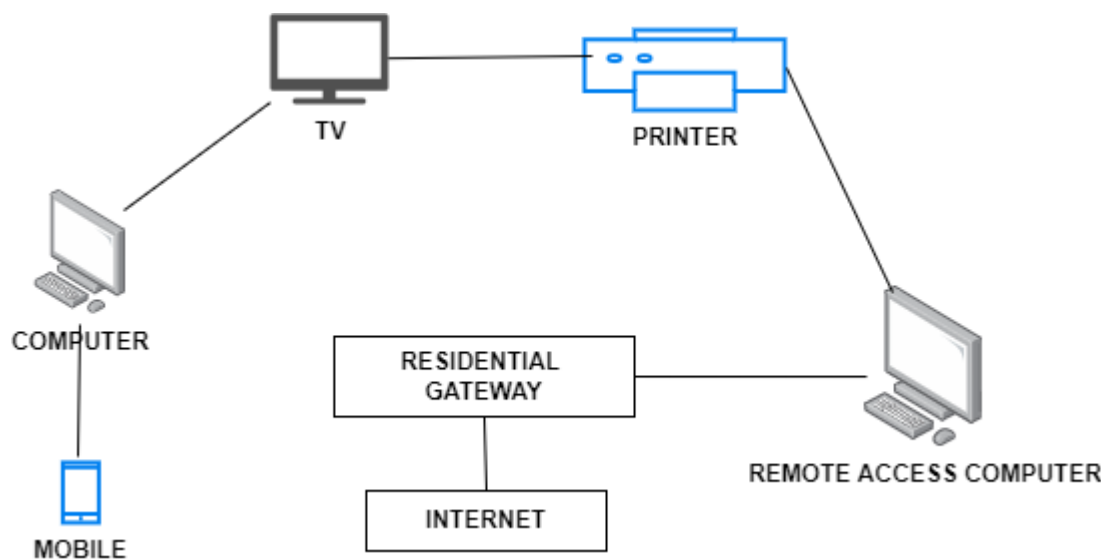


Figure 2

The inputs from satellite dish antennae, cable coaxial lines, phone lines, and wireless antennae are all placed on the front channels. A tuner is connected to the inputs. The front channels' signals are received and decoded by a decoding device. The set-top box decodes the signals it receives from the service provider. Multi-channel tuners are found in set-top boxes. The user-

selected channel is divided by a de-multiplexer. The audio and video are likewise separated. The channel's access requirements are decoded by a decoder.

The gadget has a conditional access system (CAS), which restricts access to TV channels to the window of time set by the channel's service provider. A decoder decodes the signals chosen at the CAS. These are sent to home audio and video systems through the decoder.

Let's use an illustration to demonstrate this. Electricity bills are paid based on how much energy is used in a given month. Similar to this, a set-top box keeps track of the time a channel is utilized, and the service provider bills the user according to the channels and time period they have selected.

Through backchannels, the set-top box transmits its output to wireless antennae, cable coaxial lines, and telephone lines. The outputs moving through backchannels are encrypted using a keying unit. This offers interactive TV, web browsing, and the service provider feedback channels.

The most popular programming language in a set-top box is Java. Software for encryption and decryption is executed on set-top boxes. Device administration is carried out by a software element known as a device agent on behalf of the service provider. This system works in a manner akin to a mobile phone, where the device's management and administration are handled by a server provided by the mobile service provider. Here are a few instances of this:

Similar to a Smartphone, a set-top box can perform a wide range of tasks:

- ✓ Data, media, and network processing are all capabilities of a set-top box.
- ✓ It offers a platform for multimedia games powered by Java.
- ✓ Some set-top boxes have wireless keypads that can be used to control the TV, play video games, browse the Internet, and pick, tune, and modify picture and sound quality.
- ✓ Both a CD-ROM drive and hard discs are present.
- ✓ A USB port can be used to link a set-top box to a computer or printer.

EXERCISE

1. Explain how mobile devices can interact with their surroundings.
2. What are the sub-units in a set-top box?
3. Show the working of sensor-actuator pair.
4. Write a short note on smart appliances.
5. How are sensors used in robotic systems?

3.4 AUTOMOTIVE SYSTEMS

Vehicles/Automobiles today contain sophisticated computational equipment. Since the late 1960s, there has been a significant increase in the usage of computing and processing units in car engines to enhance vehicle stability, transmission and brake functions, and driving comfort and convenience. Automotive systems have undergone a transformation during the past 10 years or so. The modern automobiles come equipped with everything from sophisticated information-oriented technology like GPS navigation, reverse sensing, and night vision to communication systems like e-mail access, voice control, traffic congestion information, smartcard security control, and collision avoidance sensors. A handful of the embedded computing systems found in autos are described in this section. **Figure 3** depicts the mobile computer architecture of a car.

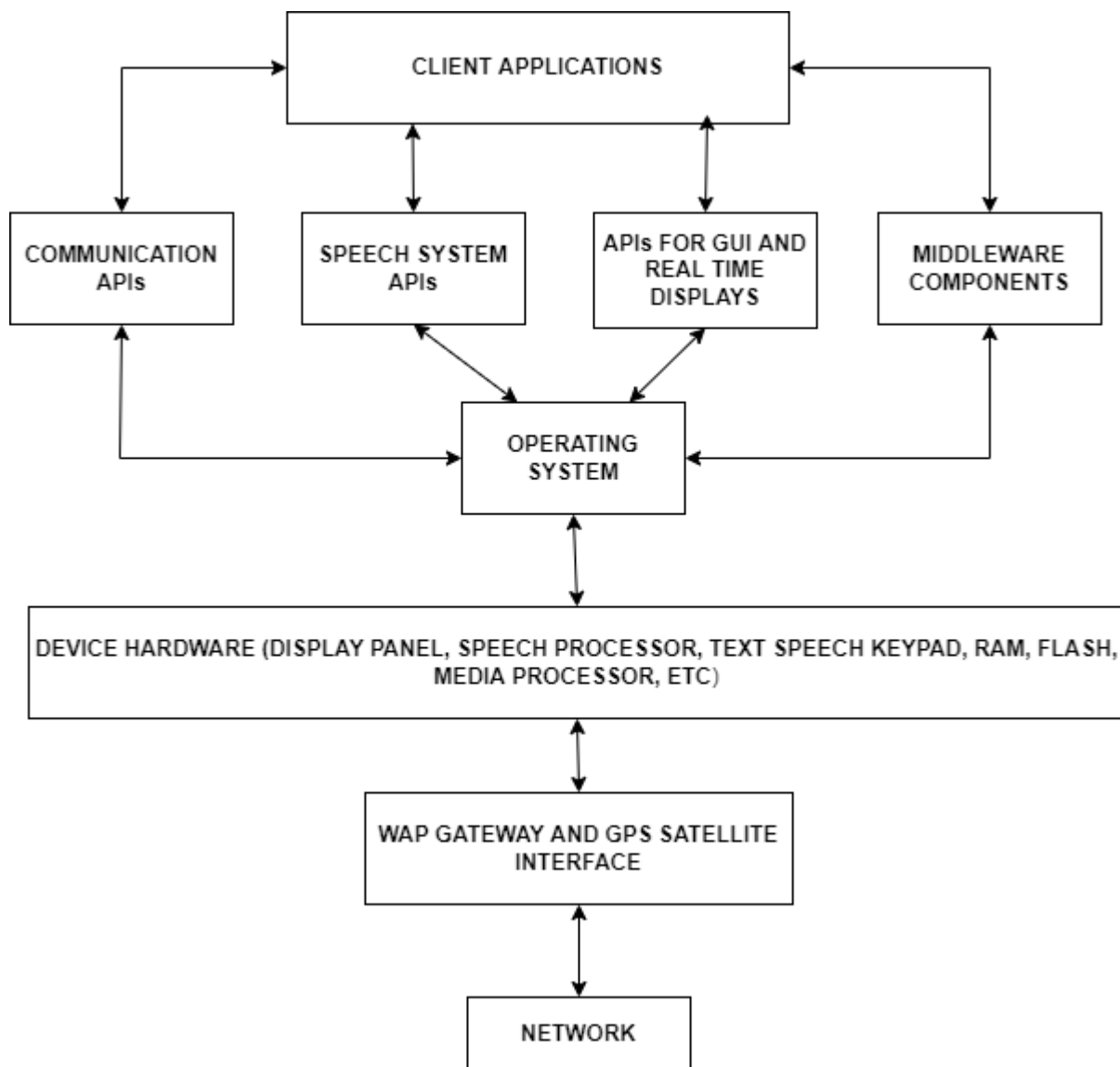


Figure 3

3.4.1 Speech Recognition System

An automobile's speech recognition system (SRS) can be programmed by an application programmer. After the SRS recognizes the voice, the driver can instruct the car to start. Application software can be set up so that the driver can provide instructions to the car to stop, keep going at the current speed, or drive within a set speed restriction. A digital signal processor is used by the SRS.

3.4.2 Messaging System

An automobile's ability to connect to the Internet is made possible by a WAP device. A service provider is able to transmit news, weather information, and stock reports in real time. WAP may be used to get road maps as well. When needed, the maps can be accessed from the memory of a computer that is integrated into a car. The GPS device and a map both aid in navigation.

Traffic reports are sent by a traffic control service. The owner of a car can sign up for a traffic control service that sends an SMS when there are delays or obstructions in traffic at different locations throughout the city. The TTS converter software then turns the messages into speech, which the driver may hear.

It allows the driver to choose routes that will allow for a quicker, obstacle-free travel.

In the event that one car approaches another too closely, an anti-collision system can alert the driver. Using a laser, infrared, or radar technology, it may also detect items that the driver cannot see. Systems that prevent collisions with other objects can also take control of the vehicle.

Application programmers can convert SMS TTS using C in Linux so that drivers don't have to take their eyes off the road to read the text on the display panel and can instead listen to the messages they've received while driving.

Java, Active Server Pages (ASP), and Java Server Pages (JSP) are further tools that programmers can use to create web-based applications and get data from databases at different portals.

An App named 'Flight' is used. As they approach the airport, the driver retrieves flight details from the airline's portal database.

3.4.3 GPS-based Navigation System

A GPS receiver can be installed in a car. It picks up signals sent by numerous GPS satellites that are in circulation around the earth.

The name of the nearby location is determined using the latitude and longitude values. The map of the immediate area is either downloaded instantly or pulled from computer storage. The position is marked on the map, which is displayed on a panel. When the car is moving, the geographic location is continuously marked and altered on a map on a display panel. It aids the

motorist in selecting the proper route that will take them there. The name of the present location and the name of the road being used can also be spoken aloud using data-to-speech converter software.

Application programmers can generate a real-time road map on the display panel in Linux or the Graphic Tool Kit (GTK) language, with the vehicle's location accurately recorded on the map. In the event that the car enters another zone, the real-time API also updates the map displayed on the screen. As the car moves, the API also continuously modifies the designated position.

3.4.4 Automobile Start and Malfunction Logins

To start the car, a smartcard or smart token can be used in its place. In addition to starting the vehicle when put into the host, the card also records data for any problems that occur while driving. The logged data and the specifics of the service history are retrieved from the card at the service workshop using a card reader connected to the computer. With this knowledge, the workshop can provide a better service. The information about the service can then be stored for later use on the card memory by the service provider's computer. The start and malfunction logging application can be created using JavaCard.

3.4.5 Sensor and Actuator Programming

There are numerous sensors and actuators in a car. For instance, warnings about tyre pressure are transmitted through pressure sensors through the display panel. Buses and the ECUs inside the car are used to connect sensors and actuators. A variety of actuators are present in the car, including those for the fuel injector, engine, seat height and angle adjusters, window screen motors, wipers, climate controllers, dashboard display panel, and entertainment systems. The ECUs (embedded systems) inside the car are connected by sensors and actuators. The CAN bus is the interface used by all computing devices.

3.4.6 Entertainment Systems

A variety of entertainment systems can be installed in a car, including FM radios and media players that can play Wave (WAV), RealAudio (RA), and MPEG-1 audio layer 3 (MP3) files. Using a WAP gateway, application developers can create programmes for downloading music from the in formats including WAV, RA, and MP3 You can download files from another system via a USB interface. Smartphone data can be downloaded via a Bluetooth device.

3.4.7 Real-time Applications Programming

An RTOS can be used by a programmer to run real-time applications on a PC connected to a car. Any of the programming languages, including Win32 API, Visual C, and Visual Basic, can be used to create applications. The OS offers APIs for networking, communication protocol, and many threads.

Real-time Java applications can also be developed using OSEK, which stands for "offene systeme und deren schnittstellen fur die elektronik in kraftfahrzeagen" (or "open systems and their interfaces for the electronics in motor vehicles"). Microcontrollers in auto engine control units use OSEK as their operating system.

3.5 LIMITATIONS AND DEVICES DESIGN CONSTRAINTS

Applications for mobile devices must adhere to design restrictions. The following are hand-held mobile device application restrictions and limitations.

3.5.1 Limitations of the Devices

Here are the wireless limitations:

- ✓ Limitations on mobility brought on by the necessity for automatic system configuration, tuning to available wireless networks, location management, network heterogeneity, portability to available band, and spectrum range of wireless medium
- ✓ consistency in bandwidth availability
- ✓ Compared to traditional LANs, wireless network connections have a longer latency interval.
- ✓ environmental barriers and transmission interference
- ✓ Security authentication required
- ✓ Security concerns with GSM, Wi-Fi, MMS, SMS, and other communication channels

3.5.2 Design Constraints for Handheld Mobile Device Applications

A designer must strike a balance between the best user experience, a long battery life, and effective wireless network connectivity. The suggested approach view of the traits typical of mobile handheld devices is presented here below.

Here are the characteristics of handheld devices as design limitations and suggested possible solution strategy.

Hand-held device characteristics as design constraints	Suggestion
--	------------

A smaller screen size that can only show a certain number of characters	Keep your attention on the user's current task. Use a straightforward layout that enables users to locate what they are looking for quickly and simply, minimises the amount of steps required for users to complete their tasks, and communicates clearly using short, clear commands and labels. Make the user (a) Informed: Make sure frequently used actions and contents, status information, and notifications are obvious and accessed easily. Use concise menus. Use alternatives for less commonly used items. Balance the amount of information on the screen. List items that you use frequently at the top. (b) Confident: The user should be aware of the status of information; for example, a "yes" checkmark indicates that a message was successfully sent. Permitting re-do and undo. Offer many ways to interact with an application. Information that is simple and clear boosts user confidence.
One screen at a time flashes up	Allow users to use the application switcher to switch between programmes. Ensure user comfort by taking the following actions: Make information and screen layouts simple to understand, utilize a minimal list style, and users should find it simple to learn an application. Both expert and unskilled consumers should be catered to in design. Minimize complexity. Make gradual disclosures.

Compared to normal LANs, wireless network connections have a longer time delay.	How quickly users receive information from transmitters over the wireless network can be impacted by longer latency times for wireless network connections. The user experience can be enhanced by using an optimized antenna technology to obtain strong connectivity and quick data transfers, even in places with poor coverage, and caching the data needed later. Software for communication should dynamically adapt to its surroundings, enabling users to stay connected in more locations and focus on what matters to them.
Run down processor speed	Although processor speeds have surpassed 1 GHz, slower processing speeds can still have an impact on how customers feel about the responsiveness of computationally demanding applications like video calling. Applications can manage processor-intensive tasks using background threads.
lower memory storage	The experience that users have with an application is also impacted by memory use. Execute-in-place threads are compatible with flash memory.
Small battery life	The user experience may be impacted by a small battery life. The less frequently a device must activate the wireless connection and the longer the battery life, the more effectively it manages data.

The limitations on design and the suggested strategy are based on BlackBerry's recommendations for application developers, although they work for the majority of devices.

EXERCISE

1. What are the limitations of mobile devices?
2. Write a note about automotive systems.
3. What are the different automotive systems?
4. What are the characteristics of handheld devices?

3.6 SUMMARY

A physical equipment, appliance, or system is activated by an actuator, a device that responds to signals from a controller or central computer. A computer may or may not be built inside an actuator. The GPS satellite constellation is used to synchronize GPS trackers, which are distributed computing systems, all across the world. It is a navigational aid that is necessary. There are numerous places on the planet where GPS satellites have been positioned in orbit. Devices equipped with GPS tracker (receivers) can pick up satellite signals and decode the delivered data to read time stamps. The receiver determines the precise geographic position by using delays of the stamped time in signals obtained from the satellites. A sensor is a device that perceives the physical environment, such as proximity to an object, temperature, pressure, light, metal, and smoke. It can be connected to a controller or computer. A computer for wireless communication could be embedded in a sensor. A complex computer-based device with the ability to process data, media, and networks is referred to as a set-top box. It utilizes Java as the programming language and connects the broadcasting service network and the home TV.

3.7 FURTHER READINGS

1. <https://medium.com/@muflorintine3/smart-sensors-and-actuators-3e5c0d37fde6>
2. <https://robotics.sjtu.edu.cn/upload/course/5/files/Robot%20Sensors%20and%20Actuators-new.pdf>
3. <https://www.sciencedirect.com/topics/computer-science/automotive-system>
4. <https://www.techtarget.com/searchcustomerexperience/definition/speech-recognition>
5. <https://www.gps.gov/systems/gps/>
6. "Mobile Computing" by Asoke K Telukder, Roopa R Yuvagal, TMH
7. FUNDAMENTALS OF MOBILE COMMUNICATION by Mehaboob Mujawar, Jafar A. Alzubi
8. Introduction to Mobile Communication S Sureshkumar, Fr. J. Janet, APS. Anandaraj
9. Wireless And Mobile Communication by Sanjeev Kumar, New Age International (P) Ltd., Publishers